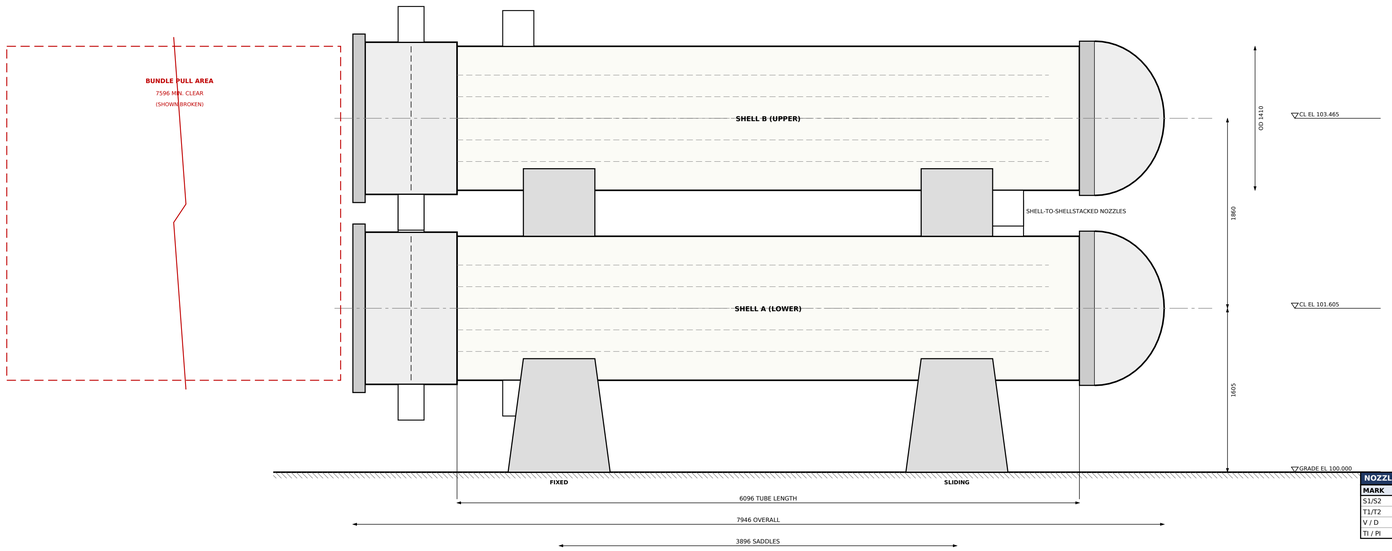
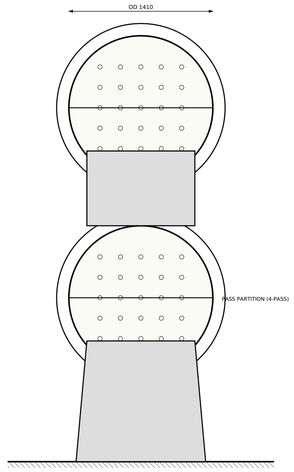
DESIGN DATA	
Item	**E-106 - Crude / Middle (kero) pumparound**
TEMA type / size	**1350-6096 AES (Ds-L mm)**
Shells	**2 (2 series x 1 parallel), stacked 2-high**
Surface	**500 m2/shell; 1001 m2 total**
Duty / U	**12.8 MW / 320 W/m2K**
Tubes	**1055 x 25.4 OD 12 BWG (2.77 mm), 6096 mm, 4 passes**
Pitch	**31.75 mm square (cleanable)**
Design P shell / tube	**45 / 20 barg**
Design T	**270 C**
Shell matl. / t	**SA-516-70 / 30 mm**
Tubes matl.	**C5 shell / 5Cr-1/2Mo tubes**
Weight empty / filled	**27.3 / 37.7 t per shell**
Code	**ASME VIII-1 / TEMA R / API 660**

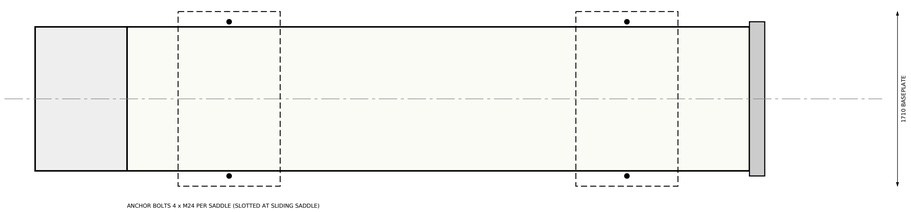
ELEVATION 1:30



END VIEW (CHANNEL END) 1:30



PLAN 1:30



NOZZLE SCHEDULE (EACH SHELL)				
MARK	**QTY**	**SIZE**	**RATING**	**SERVICE**
S1/S2	2	12"	600# RF	Shell in / out (crude)
T1/T2	2	10"	300# RF	Channel in / out (MPA)
V / D	4	1"	600# RF	Vents / drains (each shell)
TI / PI	4	1"	600# RF	Thermowell / pressure connections

AES SHELL SCHEDULE (TYPICAL ARRANGEMENT APPLIES)							
TAG	**SIZE**	**ARRGT**	**m2/SH**	**Ds mm**	**t mm**	**t/SH**	**SERVICE**
E-101	1200-6096 AES	15 x 1P	359	1200	22	19.4	Crude / Top pumparound
E-102	1300-6096 AES	15 x 1P	438	1300	25	23.6	Crude / Kerosene product
E-103	1250-6096 AES	25 x 1P	402	1250	25	21.9	Crude / LVGO pumparound/produc
E-104	1200-6096 AES	15 x 1P	378	1200	22	20.0	Crude / Diesel product
E-105	1300-6096 AES	15 x 1P	431	1300	25	23.4	Crude / Vacuum residue
E-106	1350-6096 AES	25 x 1P	500	1350	30	27.3	Crude / Middle (kero) pumparou
E-107	1150-6096 AES	15 x 1P	324	1150	30	20.0	Crude / Diesel product
E-108	1400-6096 AES	25 x 2P	549	1400	32	30.1	Crude / HVGO pumparound/produc
E-109	1000-6096 AES	15 x 1P	263	1000	25	15.5	Crude / AGO product
E-110	1300-6096 AES	15 x 2P	443	1300	32	25.7	Crude / Bottom (diesel) pumpar
E-111	1400-6096 AES	15 x 2P	547	1400	36	31.1	Crude / Vacuum residue
E-114	650-6096 AES	15 x 1P	86	650	10	5.0	Stabiliser feed / bottoms

NOTES

1. Dimensions in mm; elevations in m (grade EL 100.000).
2. Typical for AES crude preheat shells E-101..E-111; E-106 shown (2 shells in series, stacked).
3. Crude shell side (square pitch, removable bundle); hot stream tube side.
4. Bundle-pull area = tube length + 1.5 m clear at channel end.
5. Lower shell saddle fixed, upper shells sliding; intermediate supports bolted.
6. Datasheet CFU-000-ME-DS-004; weights data/mech.json.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST

100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

EXCHANGER SETTING PLAN

TYPICAL CRUDE PREHEAT EXCHANGER - TEMA AES (E-106)

DISCIPLINE	STAGE	SCALE	SHEET
MECHANICAL	FEED	1:30	1 OF 1

DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02

DRAWING NUMBER **CFU-100-ME-GA-008** REV **A**